CODEORBIT TECH

# DATA CLEANING & EXPLORATION WITH PANDAS

INTERNSHIP TASK 1

A practical data preparation report using the Titanic dataset

**Dataset:** Titanic passenger dataset (891 records, 12 columns)

**Tools:** Python, Jupyter Notebook, Pandas, NumPy, Matplotlib and Seaborn

*Prepared for internship task submission*

# 1. Executive Summary

This report documents the data cleaning and initial exploration performed on the Titanic passenger dataset using Python and Pandas. The source file contained 891 records and 12 columns. The work examined missing values, duplicate rows, data types and basic descriptive statistics, then applied documented treatments to the incomplete Age, Embarked and Cabin fields.

# 2. Dataset Overview

| Item | Description |
|---|---|
| Source file | Titanic-Dataset.csv |
| Records | 891 |
| Columns | 12 |
| Numerical columns | PassengerId, Survived, Pclass, Age, SibSp, Parch, Fare |
| Categorical columns | Name, Sex, Ticket, Cabin, Embarked |

# 3. Data Quality Assessment

The initial inspection used DataFrame preview methods, shape, info(), describe(), null counts and duplicate checks. The original dataset contained missing values in three columns and no fully duplicated rows.

| Column | Missing values | Percentage |
|---|---|---|
| Age | 177 | 19.87% |
| Cabin | 687 | 77.10% |
| Embarked | 2 | 0.22% |
| All other columns | 0 | 0.00% |

Duplicate check: 0 duplicate rows were detected. The duplicate-removal operation was retained as a validation step; the dataset remained at 891 rows and 12 columns.

# 4. Cleaning Methods Applied

**Age:** The 177 missing Age values were filled using the median age of the observed records. The median was 28.0 years. Median imputation was selected as a simple approach that is less sensitive to extreme values than the mean.

**Embarked:** The two missing Embarked values were filled using the mode, which was the most frequently occurring embarkation category (S).

**Cabin:** The 687 missing Cabin values were replaced with the text label “Unknown”. This preserves the rows and distinguishes unavailable cabin information from known cabin labels; it does not recover the original cabin numbers.

**Duplicates:** No duplicate rows were found. drop_duplicates() was applied and the index was reset; the record count did not change.

**Data types:** Column data types were inspected. The notebook identified integer, floating-point and object columns. No explicit type conversion was performed because the inspected types were retained for the analysis; this is a type-validation step rather than a documented type correction.

## 5. Descriptive Exploration

| Measure | Observed result |
|---|---|
| Mean Age after median imputation | 29.36 years |
| Median Age after cleaning | 28.00 years |
| Male passengers | 577 |
| Female passengers | 314 |
| Did not survive (Survived = 0) | 549 |
| Survived (Survived = 1) | 342 |

The value-count analysis showed more male passengers (577) than female passengers (314). The survival count was 342 survivors and 549 non-survivors. These are descriptive counts and do not, by themselves, explain the causes of survival outcomes.

## 6. Final Validation and Output

After the cleaning operations, the notebook verified that all 12 columns had zero null values and that no duplicate rows remained. The final dataset contained 891 rows and 12 columns. The cleaned data was exported as Titanic_Cleaned.csv.

| Validation item | Final result |
|---|---|
| Rows and columns | 891 × 12 |
| Total null values | 0 |
| Duplicate rows | 0 |
| Exported file | Titanic_Cleaned.csv |

## 7. Conclusion

The task established a repeatable Pandas workflow for loading, inspecting, cleaning and summarizing a structured dataset. Missing Age and Embarked values were imputed using median and mode respectively, while missing Cabin entries were labelled Unknown. Duplicate checks and final validation confirmed the resulting dataset was complete with respect to null values and contained no duplicate rows. The cleaned CSV provides the input for the subsequent EDA task.

*Scope note: Replacing missing Cabin entries with “Unknown” makes the field non-null, but the original cabin details remain unavailable. The report reflects the operations and outputs present in the submitted Task 1 notebook.*